CSCI 6360

Project 3

April 15, 2026

1 Model

```
class NN(nn.Module):
    def __init__(self, input_dim, version="m1"):
        super().__init__()

        if version == "m1":
            self.mlp = nn.Sequential(
                nn.Linear(input_dim, 64),
                nn.ReLU(),
                nn.Linear(64, 64),
                nn.ReLU(),
                nn.Linear(64, 64),
                nn.ReLU(),
                nn.Linear(64, 1)
            )
        elif version == "m2":
            self.mlp = nn.Sequential(
                nn.Linear(input_dim, 128),
                nn.ReLU(),
                nn.Linear(128, 128),
                nn.ReLU(),
                nn.Linear(128, 64),
                nn.ReLU(),
                nn.Linear(64, 1)
            )
        elif version == "m3":
            self.mlp = nn.Sequential(
                nn.Linear(input_dim, 128),
                nn.ReLU(),
                nn.Linear(128, 64),
                nn.ReLU(),
                nn.Linear(64, 32),
                nn.ReLU(),
                nn.Linear(32, 1)
            )
    def forward(self, x):
        return self.mlp(x)
```

2 In-sample regression performance on Dataset 1.

Table 1: In-sample regression performance on the Insurance Charges dataset using a learning rate of 0.01.

	$model_1$	$model_2$	$model_3$
r2	0.872	0.890	0.879
sst	196074242048.000	196074242048.000	196074242048.000
sse	25073192960.000	21539094528.000	23702245376.000
mse	18739307.145	16097977.973	17714682.643
rmse	4328.892	4012.229	4208.881
mae	2590.035	2220.552	2397.691
n	1338.000	1338.000	1338.000
aic	45448.327	45245.045	45373.092
bic	105345.214	105141.932	105269.979

Table 2: In-sample regression performance on the Insurance Charges dataset using a learning rate of 0.03.

	model_1	model_2	model_3
r2	0.896	0.943	0.899
sst	196074242048.000	196074242048.000	196074242048.000
sse	20436989952.000	11249342464.000	19720007680.000
mse	15274282.475	8407580.317	14738421.286
rmse	3908.233	2899.583	3839.065
mae	2362.465	1697.792	2307.006
n	1338.000	1338.000	1338.000
aic	45174.769	44375.934	45126.986
bic	105071.656	104272.821	105023.872

Table 3: In-sample regression performance on the Insurance Charges dataset using a learning rate of 0.05.

	model_1	model_2	model_3
r2	0.901	0.943	0.923
sst	196074242048.000	196074242048.000	196074242048.000
sse	19318460416.000	11204450304.000	15181498368.000
mse	14438311.223	8374028.628	11346411.336
rmse	3799.778	2893.791	3368.443
mae	2697.286	1704.748	2080.025
n	1338.000	1338.000	1338.000
aic	45099.459	44370.584	44777.023
bic	104996.346	104267.471	104673.910

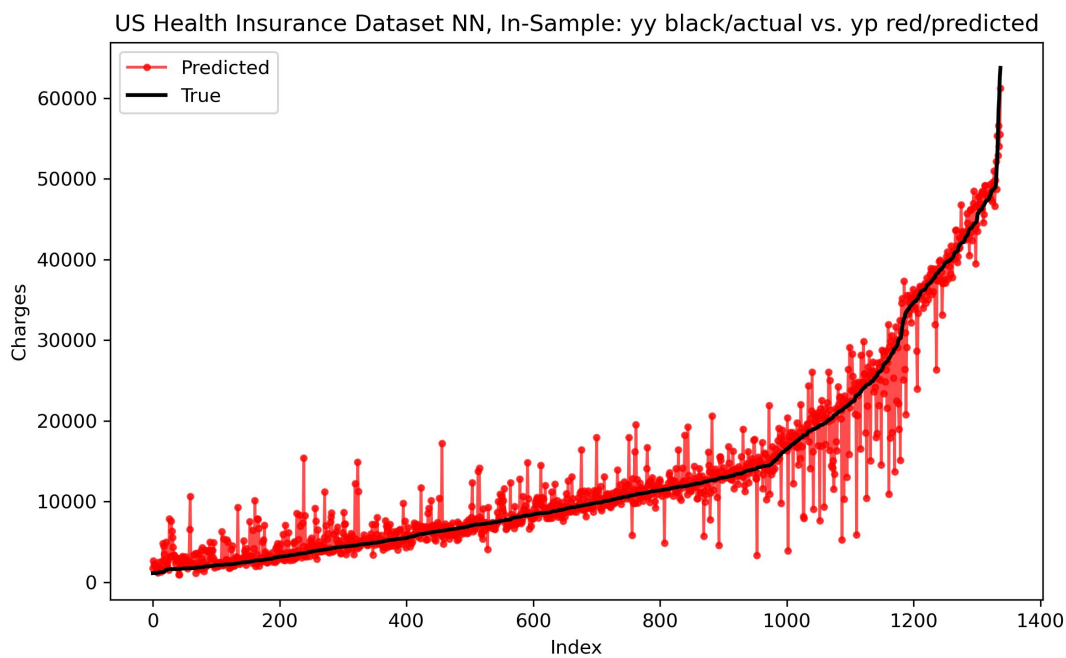

Figure 1: In-sample performance of a multi-layer neural network (Model 2) on the Insurance Charges dataset with a fine-tuned learning rate of 0.05.

3 Out-of-sample regression performance on Dataset 1

Table 4: Out-of-sample regression performance on the Insurance Charges dataset using a learning rate of 0.01.

	model_1	model_2	model_3
r2	0.844	0.857	0.847
sst	41606660096.000	41606660096.000	41606660096.000
sse	6479005184.000	5932235776.000	6378529792.000
mse	24175392.478	22135208.119	23800484.299
rmse	4916.848	4704.807	4878.574
mae	3191.396	2447.520	2853.070
n	268.000	268.000	268.000
aic	27598.227	27574.598	27594.038
bic	68969.988	68946.359	68965.799

Table 5: Out-of-sample regression performance on the Insurance Charges dataset using a learning rate of 0.03.

	model_1	model_2	model_3
r2	0.849	0.829	0.833
sst	41606660096.000	41606660096.000	41606660096.000
sse	6284380160.000	7101800448.000	6937900544.000
mse	23449179.701	26499255.403	25887688.597
rmse	4842.435	5147.743	5087.995
mae	2776.404	2935.075	2804.652
n	268.000	268.000	268.000
aic	27590.053	27622.824	27616.567
bic	68961.814	68994.585	68988.328

Table 6: Out-of-sample regression performance on the Insurance Charges dataset using a learning rate of 0.05.

	model_1	model_2	model_3
r2	0.809	0.820	0.811
sst	41606660096.000	41606660096.000	41606660096.000
sse	7938720768.000	7486139392.000	7878475776.000
mse	29622092.418	27933355.940	29397297.672
rmse	5442.618	5285.202	5421.927
mae	2930.985	2848.854	2918.611
n	268.000	268.000	268.000
aic	27652.680	27636.949	27650.639
bic	69024.441	69008.710	69022.400

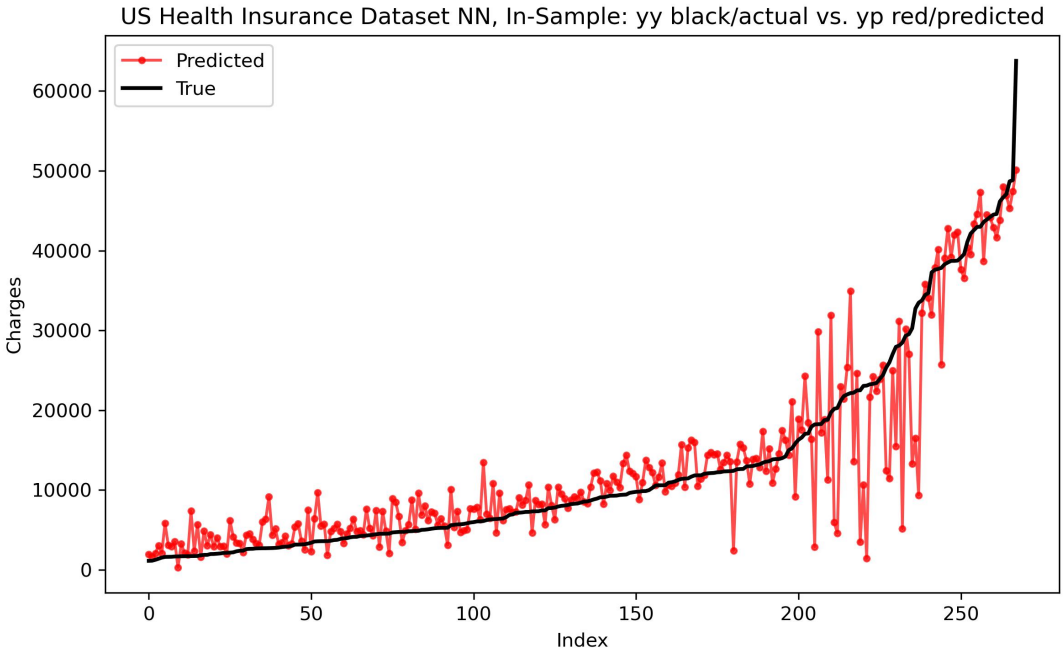

Figure 2: Out-of-sample performance of a multi-layer neural network (Model 2) on the Insurance Charges dataset with a fine-tuned learning rate of 0.01.

4 In-sample regression performance on Dataset 2

Table 7: In-sample regression performance on the Insurance Charges dataset using a learning rate of 0.01.

	model_1	model_2	model_3
r2	0.872	0.891	0.875
sst	196074225664.000	196074225664.000	196074225664.000
sse	25085812736.000	21277478912.000	24470661120.000
mse	18748738.966	15902450.607	18288984.395
rmse	4329.981	3987.788	4276.562
mae	2636.714	2235.030	2408.875
n	1338.000	1338.000	1338.000
aic	45449.000	45228.694	45415.781
bic	105345.887	105125.581	105312.668

Table 8: In-sample regression performance on the Insurance Charges dataset using a learning rate of 0.03.

	model_1	model_2	model_3
r2	0.895	0.929	0.919
sst	196074225664.000	196074225664.000	196074225664.000
sse	20505407488.000	13916065792.000	15851454464.000
mse	15325416.658	10400647.079	11847125.907
rmse	3914.769	3225.003	3441.965
mae	2219.960	2130.200	2152.952
n	1338.000	1338.000	1338.000
aic	45179.241	44660.573	44834.803
bic	105076.128	104557.459	104731.690

Table 9: In-sample regression performance on the Insurance Charges dataset using a learning rate of 0.05.

	model_1	model_2	model_3
r2	0.926	0.935	0.930
sst	196074225664.000	196074225664.000	196074225664.000
sse	14436622336.000	12748170240.000	13750960128.000
mse	10789702.792	9527780.448	10277249.722
rmse	3284.768	3086.710	3205.815
mae	1844.827	1830.518	1818.018
n	1338.000	1338.000	1338.000
aic	44709.710	44543.289	44644.603
bic	104606.596	104440.175	104541.490

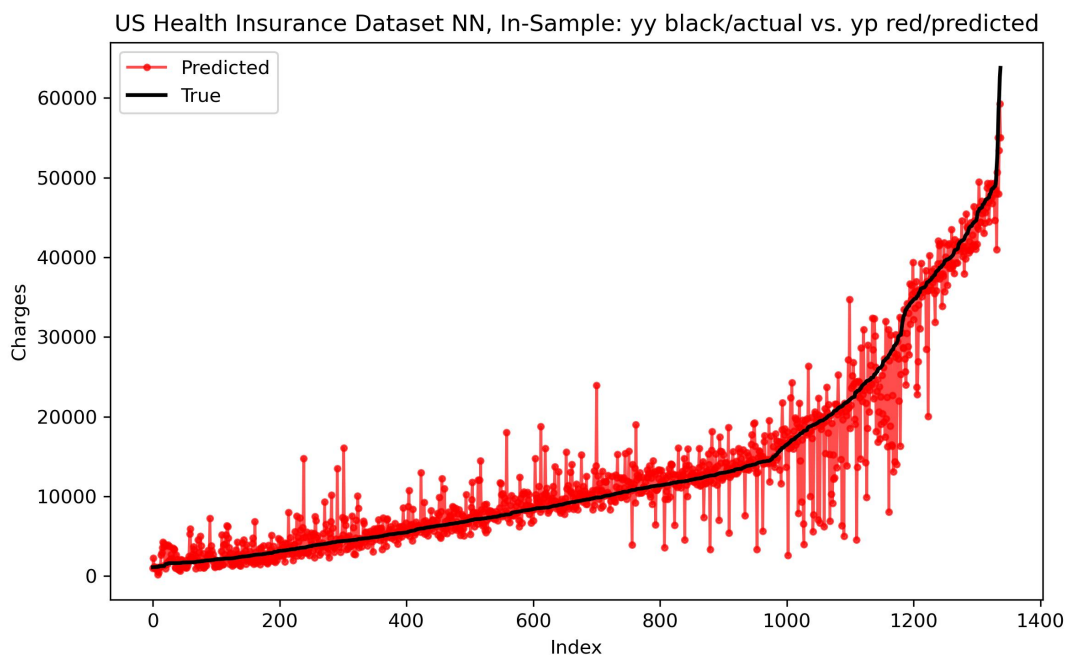

Figure 3: In-sample performance of a multi-layer neural network (Model 2) on the Insurance Charges dataset with a fine-tuned learning rate of 0.05.

5 Out-of-sample regression performance on Dataset 2

Table 10: Out-of-sample regression performance on the Insurance Charges dataset using a learning rate of 0.01.

	model_1	model_2	model_3
r2	0.855	0.846	0.845
sst	41606660096.000	41606660096.000	41606660096.000
sse	6046942208.000	6411969024.000	6435321344.000
mse	22563217.194	23925257.552	24012393.075
rmse	4750.075	4891.345	4900.244
mae	3058.983	2703.244	3008.303
n	268.000	268.000	268.000
aic	27579.731	27595.439	27596.414
bic	68951.492	68967.200	68968.175

Table 11: Out-of-sample regression performance on the Insurance Charges dataset using a learning rate of 0.03.

	model_1	model_2	model_3
r2	0.852	0.828	0.826
sst	41606660096.000	41606660096.000	41606660096.000
sse	6143384064.000	7137078784.000	7227685888.000
mse	22923074.866	26630890.985	26968977.194
rmse	4787.805	5160.513	5193.166
mae	2662.910	2758.492	2852.925
n	268.000	268.000	268.000
aic	27583.971	27624.152	27627.533
bic	68955.732	68995.913	68999.294

Table 12: Out-of-sample regression performance on the Insurance Charges dataset using a learning rate of 0.05.

	model_1	model_2	model_3
r2	0.809	0.824	0.816
sst	41606660096.000	41606660096.000	41606660096.000
sse	7933894656.000	7311921664.000	7659736576.000
mse	29604084.537	27283289.791	28581106.627
rmse	5440.964	5223.341	5346.130
mae	2974.554	2738.725	2750.858
n	268.000	268.000	268.000
aic	27652.517	27630.638	27643.093
bic	69024.278	69002.399	69014.854

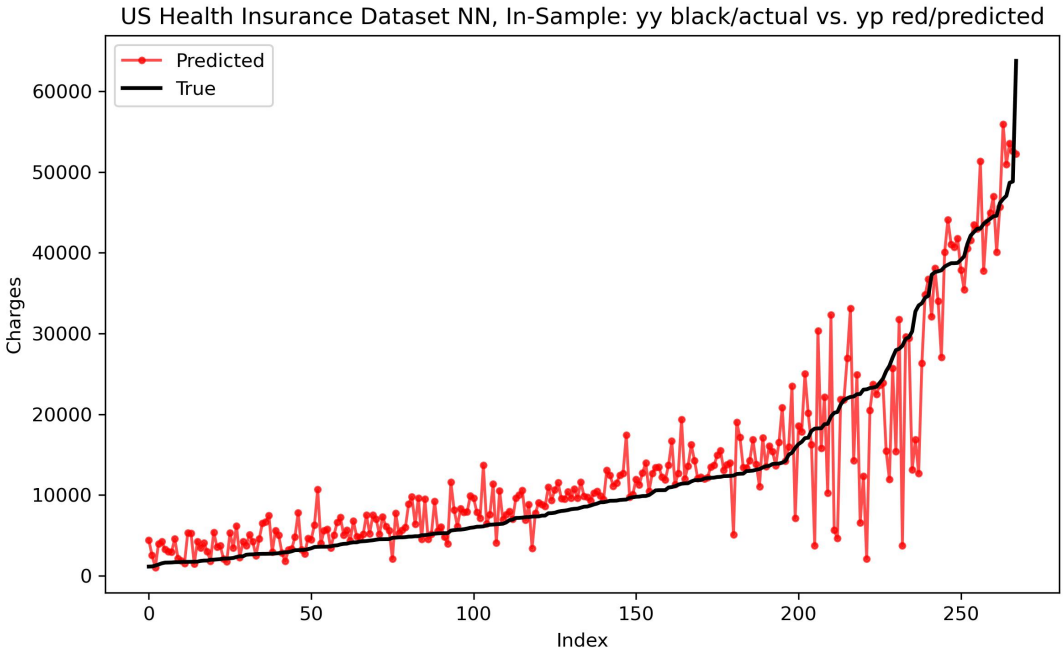

Figure 4: Out-of-sample performance of a multi-layer neural network (Model 1) on the Insurance Charges dataset with a fine-tuned learning rate of 0.01.